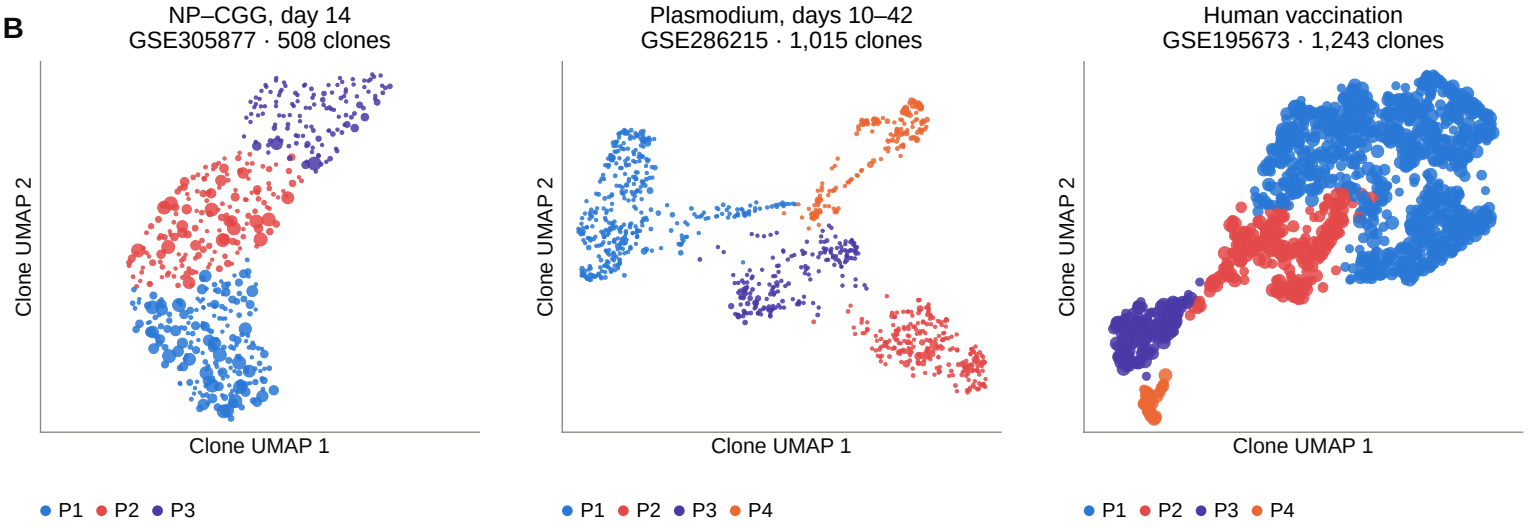
A

Model antigen	Cell types	Figure
NP–CGG · GSE305877 WT day 14 · 1 mouse · 13,727 cells · NP+ B220+ sort	GC · PB · PC · Mem	B; S1 (1)
Oral OVA + CT · GSE284622 3 reporter mice · 16,619 cells · GC-fate sort (tdTomato)	GC · PB · PC · Mem	Fig. 3
NP–OVA · GSE287123 7 mice · 10,109 cells · 36 h division reporter	GC · PB · PC · Mem	S1 (2)
RBD protein / mRNA · GSE287123 10 mice · 36,188 cells · division, bait OR LZ/DZ gates	GC · PB · PC · Mem	Fig. 4; S1 (2)
NP–OVA · GSE246382 11 mice · 884 cells · day 14 · Smart-seq2 · GC/PC gates; W33L	GC · PB · PC · Mem	S1 (1)
Plasmodium infection		
P. chabaudi AS · experiment 1 20 mice · 40,623 cells · days 0–14 · spleen	GC · PB · PC · Mem	—
P. chabaudi AS · experiment 2 · GSE286215 42 mice · 72,359 cells · days 10–42 · treated / control / naive	GC · PB · PC · Mem	B–D
Human vaccination		
BNT162b2 · GSE195673 8 participants · 193,442 cells · days 28–201 · LN FNA + blood	GC · PB · PC · Mem	B; S2; S6

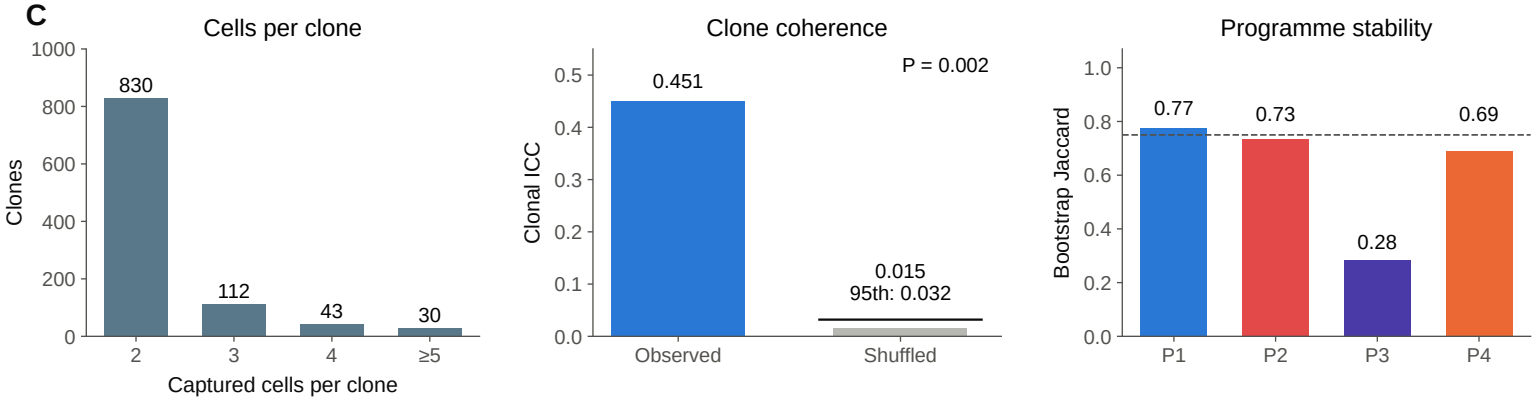
Paired scRNA-seq + BCR data. Coloured boxes mark sampled cell types.

B



Colours identify programmes within each dataset. Point area shows captured clone size.

C

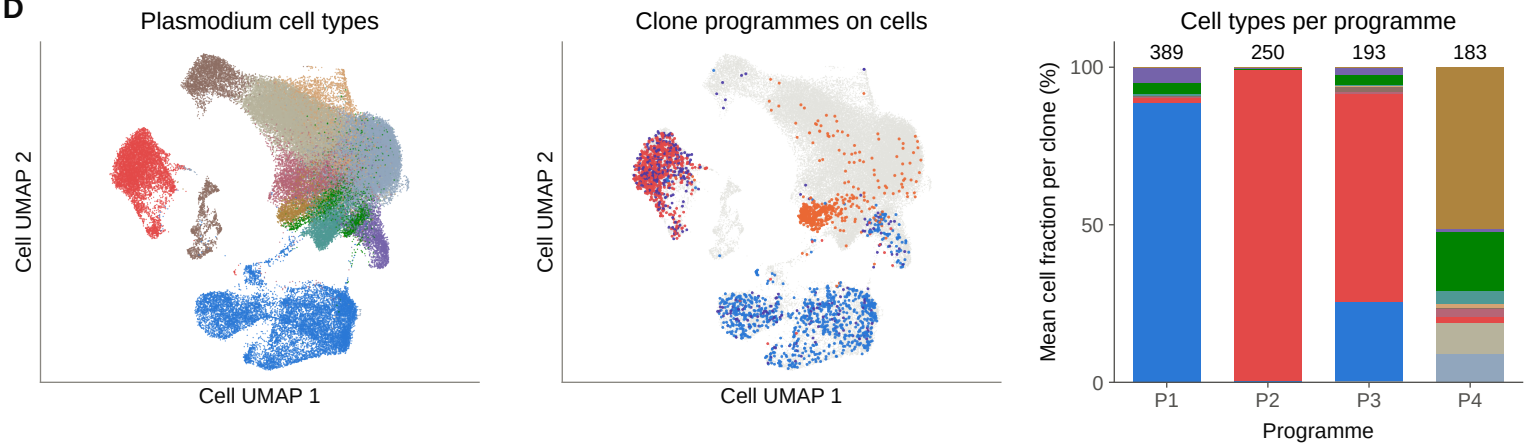


1,015 profiles; 830 contain two cells.

500 shuffles within each mouse.

50 resamples; dashed line = 0.75.

D



2,350 / 72,359 cells mapped (3.25%); grey = other cells. Bar labels = clone counts.